Task 8: Critical Evaluation of Bayesian Statistical Methods

8.1 Naïve Bayes

What it is and why we tested it here

Naïve Bayes is a probabilistic classifier built directly on Bayes' theorem, with the simplifying ("naïve") assumption that predictors are conditionally independent given the class. It is widely used in medical diagnosis and disease prediction - the literature reports Naïve Bayes performing competitively, and sometimes surprisingly well, against more complex classifiers in domains such as heart disease prediction, breast cancer diagnosis, and lung cancer survivability, and it has been specifically applied to predicting severe disease progression from clinical and laboratory parameters in a resource-constrained public hospital setting (a Mexican case-control study predicting progression to severe dengue in children, achieving 78% sensitivity and 91% specificity from routine clinical/lab variables) - a context genuinely comparable to ours.

Critically, we already know from Tasks 3 and 5 that our own predictor set violates Naïve Bayes' core independence assumption: MCH and MCV are correlated at r=0.88, HB and PCV at r=0.96, and MCV carries a VIF of 461 in the equivalent logistic regression. We therefore treat this as a deliberate stress-test of Naïve Bayes under known, quantified assumption violation, rather than assuming it would perform well.

Results

Model	Accuracy	Sensitivity	Specificity	Precision	AUC (approx.)
Naïve Bayes	88.3%	87.4%	89.0%	86.0%	0.947
Logistic Regression (same predictors, for comparison)	92.5%	91.6%	93.1%	91.1%	0.983

Interpretation

Naïve Bayes underperforms logistic regression by roughly 4 percentage points of accuracy and a meaningfully lower AUC, which is consistent with - and a direct empirical confirmation of the independence-violation concern. This mirrors the wider literature's finding that Naïve Bayes classifiers are known to be sensitive to interdependencies across features, and reflects a genuine trade-off documented across many medical classification comparisons: Naïve Bayes is not always the top performer, but its simplicity and fairly competitive performance make it a reasonable, fast baseline even when it is not the best choice. Notably, Naïve Bayes still achieves respectable absolute performance (88.3% accuracy) despite the violated assumption - a reasonably graceful degradation rather than a catastrophic failure, which is itself a useful, evidence-based finding for the client.

Practical implications

Naïve Bayes is not recommended as the client's primary screening model given the measured accuracy gap, but its computational simplicity and reasonable performance despite assumption violation make it a viable lightweight fallback for instance, in a low-resource setting where a simple, fast, easily explainable rule is preferable to a more complex model, consistent with the resource constraints on Sri Lankan primary care laboratories documented in Task 1.

8.2 Bayesian Regression

What it is and why we tested it here

Bayesian regression treats model coefficients as random variables with a prior distribution, updated via the data's likelihood to produce a posterior distribution providing not just point estimates but a full quantification of uncertainty, and allowing external knowledge to be incorporated through the prior. Bayesian methods are widely reported in the clinical research literature as advantageous specifically for small sample sizes (rare diseases, sparse subgroups) and for incorporating prior knowledge from previous studies or expert judgement - both directly relevant to our project, given Institution_A's small historical sample (n=710) and the substantial existing literature on anemia risk factors that a fully "blank slate" frequentist model ignores.

A genuine technical obstacle we encountered - and how it was resolved

Our first attempt used full MCMC sampling (MCMCpack::MCMClogit) with a standard weakly informative Normal prior. This failed: the Metropolis sampler achieved a 0% acceptance rate across 12,000 iterations, regardless of proposal tuning, prior tightness, or subsample size. We diagnosed this as a consequence of near-perfect class separation in our predictor set - the same phenomenon underlying Task 5's high accuracy and the recurring "fitted probabilities numerically 0 or 1" warning from R's own glm() - which creates an extremely peaked likelihood surface that a basic random-walk sampler cannot traverse.

This is a genuine, documented limitation of naïve MCMC approaches to Bayesian logistic regression under separation, not a hypothetical concern. We resolved it using the published solution to exactly this problem: a weakly informative Cauchy (0, 2.5) prior on the coefficients (Gelman, Jakulin, Pittau & Su, 2008), implemented via arm::bayesglm, which regularizes coefficients away from the separation boundary using a fast, well-behaved approximate posterior mode estimate.

Results

Term	Frequentist MLE (SE)	Bayesian posterior mode, Cauchy prior (SE)
age_at_test	0.0701 (0.00187)	0.0699 (0.00186)
sexMale	2.10 (0.080)	2.09 (0.080)
MCH	−0.706 (0.274)	−0.486 (0.221)
MCHC	−1.04 (0.236)	−1.23 (0.192)
MCV	0.448 (0.106)	0.349 (0.086)
PCV	−1.82 (0.087)	−1.76 (0.081)
RBC	3.19 (0.699)	2.68 (0.645)

Test-set predictive performance: **Accuracy 92.50%, Sensitivity 91.64%, Specificity 93.16%, AUC ≈ 0.983** - essentially identical to (marginally better than) the frequentist logistic regression (92.45% accuracy).

Interpretation

Two findings stand out. First, coefficients that were already well-estimated with small standard errors (age, sex) are almost unchanged between the two approaches exactly as expected, since with a large sample and a weak prior, the likelihood dominates. Second, the coefficients most affected by multicollinearity (RBC: VIF=59.3, MCV: VIF=461) show the largest shrinkage toward zero and the tightest resulting standard errors under the Bayesian approach - the Cauchy prior is doing real, visible regularizing work precisely where the data is least informative on its own. This is a clean, dataset-grounded illustration of a textbook Bayesian advantage: the prior stabilizes estimation exactly where the likelihood alone is weak or unstable, without materially changing predictions where the data already speaks clearly.

Practical implications

For this client, Bayesian regression with a weakly informative prior offers a genuine, measured advantage over ordinary logistic regression: more stable, less collinearity-distorted coefficients, at effectively no cost in predictive accuracy. This is a lighter-weight alternative to fully regularized regression (Ridge/LASSO/Elastic Net from Task 5) that additionally provides a principled uncertainty quantification (credible intervals) rather than only point estimates and confidence intervals derived from asymptotic approximations.

8.3 Bayesian Decision Making

What it is and why it matters for this project

Bayesian decision theory formalizes decision-making under uncertainty by choosing the action that minimizes expected loss (or maximizes expected utility), given the posterior probability of each possible outcome and an explicit statement of the costs involved. This is distinct from simply reporting a probability - it is about what to do given that probability, which is precisely the gap between Task 5's predictive model (which outputs a probability) and an actual operational screening decision (flag this patient, or not). The FDA's own guidance on Bayesian statistics in medical device trials explicitly identifies "optimal decision making (Bayesian decision theory)" as one of the technique's principal real-world applications, and a broader

review of Bayesian methods in healthcare system assessment highlights their use for producing "decision-relevant outputs" in resource allocation and policy appraisal specifically.

What we did

Every model built so far (Task 5, and Section 8.2 above) has used the conventional default classification threshold of 0.5 - implicitly treating a false negative (telling a patient they are not anemic when they are) and a false positive (flagging a non-anemic patient for unnecessary follow-up) as equally costly. In a clinical screening context this is rarely defensible: missing a true case delays treatment and risks the downstream harms of untreated anemia, while a false positive typically triggers only a low-cost confirmatory test. We illustrate Bayesian decision-making by finding the threshold that minimizes total expected loss, using the Bayesian model's posterior predictive probabilities from Section 8.2, under an explicitly stated cost ratio: missing a true case costs 5× as much as a false alarm.

Results

Threshold	False Negatives	False Positives	Total loss
0.500 (conventional default)	246	261	1,491
0.150 (expected-loss-optimal)	34	658	**828**

Shifting the decision threshold from 0.5 to the loss-minimizing 0.15 reduces total expected loss by 44.5% on the held-out test set - at the cost of substantially more false positives (658 vs. 261), but a large reduction in the clinically more serious false negatives (34 vs. 246).

Interpretation

This is not a marginal, theoretical benefit - it is a substantial, quantified improvement achievable without retraining the model at all, purely by making the classification decision rule explicitly reflects the true asymmetric costs of the two error types, rather than defaulting to an arbitrary 0.5 cutoff inherited from generic classification convention. This is Bayesian decision theory's core contribution: separating "what is the probability of anemia" (a statistical question, answered

similarly by frequentist or Bayesian models) from "given that probability and our costs, what should we do" (a decision-theoretic question that neither frequentist point estimation nor an arbitrary default threshold actually answers).

Practical implications

We recommend the client engage clinical and health-economic stakeholders to establish a defensible, locally appropriate cost ratio (our 5:1 assumption is illustrative, not empirically derived from Sri Lankan treatment/follow-up costs) and then set the operational screening threshold using this expected-loss framework, rather than a default cutoff. This directly operationalizes the "screening" framing of this project's sub-topic - a screening tool's entire value proposition rests on correctly calibrating the trade-off between missed cases and unnecessary follow-ups for the specific resource constraints.

8.4 When Bayesian Methods May Outperform Traditional Approaches

Drawing on both the literature and our own results:

- **Small or sparse subgroups**: Bayesian methods are repeatedly identified in the literature as most advantageous for small sample sizes and rare-disease/rare-subgroup contexts, where frequentist inference may struggle. This is directly relevant to Institution_A (n=710) in our own dataset - a future institution-specific Bayesian analysis could borrow strength from the pooled/other-institution data via a hierarchical (partial-pooling) model, an approach explicitly highlighted in the healthcare Bayesian-methods literature as providing more stable estimates in sparse settings.
- **Incorporating existing knowledge**: unlike a frequentist analysis, which treats every new dataset as if nothing were previously known, Bayesian priors let the client formally incorporate Sri Lanka's existing anemia epidemiology literature into future analyses - reducing the sample size needed to reach a confident conclusion, since the prior itself contributes information equivalent to additional observations.
- **Direct probability statements for decision-making**: a Bayesian credible interval supports the direct statement "there is a 95% probability the true

effect lies in this range" - a more intuitive and directly decision-relevant statement than a frequentist confidence interval's more convoluted long-run interpretation, which matters when communicating findings to non-statistician hospital administrators or Board members.

- **Numerically unstable frequentist estimation**: our own Section 8.2 result is a direct empirical example - the Bayesian (Cauchy-prior) coefficients for the most collinear variables were both more stable and came with more defensible uncertainty estimates than the frequentist MLE under near-separation.
- **Where frequentist methods are comparable**: it is equally important to note the literature's finding that all three approaches (frequentist logistic regression, Naïve Bayes, and automated ML) performed similarly well in predicting mental health outcomes in a large cohort study, and our own results likewise show Bayesian regression achieving performance nearly identical to frequentist logistic regression once separation is properly handled - Bayesian methods are not a universal upgrade, but a genuinely superior choice in specific, identifiable circumstances (small samples, unstable estimation, need for formal decision rules), which this task's brief specifically singles out as the crux of the critical evaluation the client wants.

8.5 Limitations

- **Naïve Bayes' independence assumption is measurably violated** in this dataset, and this cost ~4 percentage points of accuracy relative to logistic regression.
- **Standard MCMC-based Bayesian regression failed outright** on our predictor set due to near-perfect separation, requiring a different (Cauchy-prior, approximate) implementation. Naïve application of "the Bayesian approach" is not automatically robust and requires the same statistical judgement and troubleshooting as any other method.
- **Prior specification introduces subjectivity.** Our illustrative 5:1 cost ratio in Section 8.3 was chosen for demonstration, not derived from actual Sri Lankan clinical/economic data - a real prior-elicitation exercise with the

client's clinical stakeholders would be needed before operational deployment, and the literature explicitly flags that results remain sensitive to prior specification, stressing the need for sensitivity analysis.

- **Computational cost**: proper Bayesian methods (especially full MCMC) are typically more computationally demanding than frequentist equivalents - a real, if usually manageable, practical consideration for a resource-constrained analytics team.

8.6 Overall Recommendation

We recommend the client consider Bayesian methods selectively, not as a wholesale replacement for the frequentist/regularized:

- **Naïve Bayes**: not recommended as the primary model given its measured accuracy gap, but viable as an extremely lightweight fallback.
- **Bayesian regression (Cauchy prior)**: a genuinely attractive alternative to Task 5's regularized models for future work, particularly once the client wants formal credible intervals or wishes to incorporate prior literature/expert knowledge into the model directly.
- **Bayesian decision theory**: the strongest, most immediately actionable recommendation from this task - the client should adopt an expected-loss framework for setting the operational screening threshold, once appropriate cost ratios are established with clinical input, rather than defaulting to an arbitrary 0.5 cutoff. Our illustration shows this alone could reduce total expected error cost by over 40%.

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